Supplementary Information

A Probabilistic Ensemble-Based Bayesian Method for Mechanistic Inference of Complex Systems

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S1 Likelihood Convexity

To determine whether the likelihood is convex or not with respect to the interaction matrix weights, we compute the Hessian matrix and verify that it is semi-positive definite.

We focus on a single node i. Under the Euler-Maruyama discretization, the objective function to minimize (the negative log-likelihood) over M time steps is the sum of squared residuals:

$$\mathcal{L}(\mathbf{W}_i) = \frac{1}{4D\Delta t}\sum_{k=1}^{M}(e_{i,k})^2$$

Where the residual error $e_{i,k}$ at time step k is defined as:

$$e_{i,k} = x_{i,k} - x_{i,k-1} - \Delta t\Big(- \gamma_i x_{i,k-1} + V_i\sigma(z_{i,k-1})\Big)$$

And $z_{i,k-1} = \sum_j W_{ij}x_{j,k-1}$ is the total input to the sigmoid function.
The first derivative is:

$$\frac{\partial\mathcal{L}}{\partial W_{ij}} = \frac{1}{2D\Delta t}\sum_{k=1}^{M} e_{i,k}\frac{\partial e_{i,k}}{\partial W_{ij}}$$

Taking the derivative of the residual $e_{i,k}$ with respect to W_{ij}:

$$\frac{\partial e_{i,k}}{\partial W_{ij}} = -\Delta t\, V_i\, \sigma'(z_{i,k-1})\, x_{j,k-1}$$

So the gradient is:

$$\frac{\partial\mathcal{L}}{\partial W_{ij}} = -\frac{V_i}{2D}\sum_{k=1}^{M} e_{i,k}\, \sigma'(z_{i,k-1})\, x_{j,k-1}$$

Taking the second derivative, we get the entries of the Hessian matrix:

$$H_{i,j} = \frac{\partial^2\mathcal{L}}{\partial W_{ij}\partial W_{il}} = \frac{1}{2D\Delta t}\sum_{k=1}^{M}\left[\frac{\partial e_{i,k}}{\partial W_{ij}}\frac{\partial e_{i,k}}{\partial W_{il}} + e_{i,k}\frac{\partial^2 e_{i,k}}{\partial W_{ij}\partial W_{il}}\right]$$

Unpacking each term:

$$\frac{\partial e_{i,k}}{\partial W_{ij}}\frac{\partial e_{i,k}}{\partial W_{il}} = \Delta t^2 V_i^2[\sigma'(z_{i,k-1})]^2\, x_{j,k-1}\, x_{l,k-1}$$

$$e_{i,k}\frac{\partial^2 e_{i,k}}{\partial W_{ij}\partial W_{il}} = -\Delta t\, V_i\, e_{i,k}\, \sigma''(z_{i,k-1})\, x_{j,k-1}\, x_{l,k-1}$$

Combining both terms and factoring out $x_{j,k-1}x_{l,k-1}$, the full Hessian matrix element is:

$$H_{jl} = \frac{V_i}{2D} \sum_{k=1}^{M} x_{j,k-1} x_{l,k-1} \underbrace{\left[\Delta t \, V_i \big(\sigma'(z_{i,k-1})\big)^2 - e_{i,k} \, \sigma''(z_{i,k-1}) \right]}_{c_k}$$

In matrix form, the Hessian is $\mathbf{H} = \mathbf{X}^T \mathbf{C} \mathbf{X}$, where $\mathbf{C}$ is a diagonal matrix containing the coefficients c_k.

$$c_k = \Delta t \, V_i \big(\sigma'(z_{i,k-1})\big)^2 - e_{i,k} \, \sigma''(z_{i,k-1}) \tag{S1}$$

For the likelihood $\mathcal{L}$ to be strictly convex, the Hessian $\mathbf{H}$ must be Positive Semi-Definite (PSD) everywhere. This requires that the coefficient $c_k \geq 0$ for all time steps k. Looking at S1, we can notice that the first term is always positive, while the second one can be negative, given that the second derivative of the sigmoid changes sign with respect to the sign of the argument.

This fact does not allow to be sure that a simple gradient descent algorithm can safely converge to the global minimum of the likelihood function.

S2 Parameter Indistinguishability

S2.1 Equilibrium Scaling Degeneracy ($\gamma_i \sim V_i$)

Near dynamic equilibrium or steady state, the drift vanishes ($\Delta x_i(t)/\Delta t \approx 0$), yielding the stationary expectation:

$$x_i^* = \frac{V_i}{\gamma_i} \sigma \left(\sum_j W_{ij} x_j^* \right)$$

If both degradation γ_i and transcription scale V_i are multiplied by a scale factor $c > 0$:

$$\gamma_i \mapsto c\gamma_i, \quad V_i \mapsto cV_i \implies \frac{cV_i}{c\gamma_i} = \frac{V_i}{\gamma_i}$$

Thus, the stationary expression x_i^* is invariant to uniform scaling of (γ_i, V_i). In data lacking strong non-equilibrium transients, MCMC proposals along the ridge $\gamma_i = cV_i$ produce identical steady-state predictions.

S2.2 Sigmoid Saturation Degeneracy ($|m_i| \gg 1$)

The derivative of the drift term with respect to an edge weight W_{ij} is:

$$\frac{\partial f_i}{\partial W_{ij}} = V_i \frac{\partial \sigma(m_i)}{\partial m_i} x_j = V_i \sigma(m_i) \left(1 - \sigma(m_i)\right) x_j$$

When total regulatory input $|m_i| = \left| \sum_j W_{ij} x_j \right|$ is large, the sigmoid saturates ($\sigma(m_i) \to 0$ or 1). In this saturated regime, the gradient vanishes:

$$\lim_{|m_i| \to \infty} \sigma(m_i)(1 - \sigma(m_i)) = 0$$

Consequently, once an edge W_{ij} drives node i into saturation, further changes to W_{ij} or addition of new edges W_{ik} produce virtually zero change in log-likelihood ($\Delta S \approx 0$), causing edge misinference.

S2.3 Multiplicative Linear Indeterminacy ($V_i W_{ij} = \text{const}$)

When regulatory inputs are small ($|m_i| \lesssim 1$), the sigmoid operates in its linear regime ($\sigma(m_i) \approx \frac{1}{2} + \frac{1}{4} m_i$). The drift simplifies to:

$$f_i(x) \approx -\gamma_i x_i + \frac{1}{2} V_i + \frac{1}{4} V_i \sum_j W_{ij} x_j$$

In this linear regime, scaling $V_i \mapsto \alpha V_i$ while inversely scaling edge weights $W_{ij} \mapsto \alpha^{-1} W_{ij}$ leaves the effective interaction strength $V_i W_{ij}$ unchanged, introducing a multiplicative degeneracy between node parameters and edge weights.

S3 Exact Analytical Transition Moments (Itô Isometry)

The Euler-Maruyama discretization, while computationally simple, is mathematically guaranteed to become unstable at large measurement intervals (Δt), failing to capture the exponential relaxation of the degradation term and incorrectly modeling the variance. To overcome this, we derive the exact analytical transition moments under a zero-order hold assumption, where the regulatory input $m_i(t) = \sum W_{ij} x_j(t)$ is assumed constant over the interval Δt.

Under this assumption, the SDE for node i reduces to a strictly linear CIR process:

$$dx_t = (-\gamma x_t + b)dt + \sqrt{2D x_t} dW_t \tag{S2}$$

where $b = V\sigma(m_i(t))$. We define the steady-state asymptote as $\mu_\infty = b/\gamma$.

The exact expected mean $\mathbb{E}[x_{t+\Delta t} \mid x_t]$ is the solution to the deterministic ODE part:

$$\mu(t + \Delta t) = \mu_\infty + (x_t - \mu_\infty) e^{-\gamma \Delta t} \tag{S3}$$

To find the exact accumulated variance, we apply Itô's Isometry. The variance of the integrated stochastic process over $[0, \Delta t]$ is:

$$\mathrm{Var}(x_{t+\Delta t} \mid x_t) = \mathbb{E}\left[\int_0^{\Delta t} e^{-2\gamma(\Delta t - s)} 2D \cdot \mathbb{E}[x_s \mid x_t] ds \right] \tag{S4}$$

Substituting the exact expected mean $\mathbb{E}[x_s \mid x_t]$ into the integral and evaluating it yields the exact closed-form solution:

$$\mathrm{Var} = \frac{2D}{\gamma}(x_t - \mu_\infty)(e^{-\gamma \Delta t} - e^{-2\gamma \Delta t}) + \frac{D \mu_\infty}{\gamma}(1 - e^{-2\gamma \Delta t}) \tag{S5}$$

This formula correctly collapses to the Euler-Maruyama variance $2D x_t \Delta t$ as $\Delta t \to 0$, and saturates to the finite steady-state variance $D\mu_\infty/\gamma$ as $\Delta t \to \infty$.

S4 Evaluation Metrics Formulation

Because the inference produces a posterior ensemble rather than a single point-estimate, we evaluate the structural recovery using probabilistic metrics based on the marginal edge probability p_{ij}, which is the fraction of MCMC sweeps in which the edge $i \to j$ exists. The Area Under the Receiver Operating Characteristic (AUROC) and Precision-Recall Curve (AUPRC) were calculated by thresholding p_{ij} directly, bypassing binary MAP predictions.

For the perturbation analysis, where the goal is to predict the downstream steady-state x_j after a knockout or over-expression, we evaluate the ensemble's predictive power using a **Kernel Density Estimation (KDE) log-likelihood**. We construct a probability density function $f(x)$ from the ensemble of predicted steady-states using a Gaussian kernel, and evaluate the likelihood of the true systemic steady-state x^{true} under this density:

$$\mathrm{LL}_{KDE} = \log f(x^{true}) = \log \left(\frac{1}{S} \sum_{s=1}^{S} \mathcal{N}(x^{true} \mid x^{(s)}, h^2) \right) \tag{S6}$$

where S is the number of ensemble samples and h is the bandwidth chosen via Scott's rule.

S5 Baseline Competitor Hyperparameters

To ensure a fair and reproducible benchmarking study against standard inference paradigms, the baseline methods were executed using the following settings:

- **dynGENIE3:** Executed using the default Random Forest implementation tailored for time-series data. The number of trees was set to 1000 to ensure stable variable importance metrics.
- **Sparse ODE via Lasso (Inferelator-style):** Implemented as a regularized linear regression problem predicting the finite-difference derivative $\Delta x_i/\Delta t$. The L_1 regularization penalty (α) was optimized independently for each node using 5-fold cross-validation (LassoCV) to enforce sparsity.
- **PySINDy:** The Sequential Thresholded Least Squares (STLSQ) optimizer was used. To remain comparable with our SDE model's capacity, the feature library was restricted to first-order polynomials (linear combinations of the node states). The sparsity threshold was optimized via cross-validation.
- **Granger Causality:** Executed using a classical Vector Autoregression (VAR) model. The optimal maximum lag was selected using the Akaike Information Criterion (AIC). F-tests were performed to determine the statistical significance of predictive causality, and the resulting p-values ($-\log_{10}(p)$) were used as confidence scores for the AUPRC evaluation.
- **Elastic Net VAR:** Implemented as an extension to standard Lasso using Scikit-Learn's `ElasticNet`. It combined L_1 and L_2 penalties, fitting independent regression models mapping $\mathbf{X}(t)$ to $\frac{d\mathbf{X}}{dt}$. The resulting coefficients served directly as the inferred continuous causal interaction matrix W.
- **Gaussian Process ODEs (GP-ODEs):** Fit using Gaussian Process regressors with an Automatic Relevance Determination (ARD) Radial Basis Function (RBF) kernel. The inverse of the learned length scales for each input dimension was extracted as a proxy for the causal interaction strength.
- **Neural ODEs:** Implemented via PyTorch using a Multi-Layer Perceptron (MLP) with a single hidden layer to map $\mathbf{X}(t)$ to $\frac{d\mathbf{X}}{dt}$ via Mean Squared Error minimization over 200 epochs. To extract a linear causal skeleton, the Jacobian of the learned function $\frac{\partial f_i}{\partial X_j}$ was evaluated at the empirical mean of the training data.
- **PCMCI:** Implemented via the `tigramite` library. Simulated transient pairs were concatenated to form sequential transitions. The resulting p-value matrices for lagged links ($\tau = 1$) via Partial Correlation tests were converted to continuous confidence scores ($1 - p$).
- **Dynamic Bayesian Networks (DBNs):** Discretized the continuous states into equal-frequency bins and utilized `pgmpy`. Structure learning was performed across time-slices t and $t + 1$ using Hill Climb Search optimized via the Bayesian Information Criterion (BIC), blacklisting contemporaneous and backward-in-time edges.

S6 Erdős-Rényi Grid Search Benchmark

To ensure the scalability of our Bayesian framework was not strictly dependent on the presence of topological hubs, we replicated the comprehensive grid search on homogeneously random Erdős-Rényi (ER) networks (Fig. S1). For small networks ($N \leq 20$), the ER topology enabled superior quantitative parameter inference (Spearman ≈ 0.86, compared to 0.76 for Scale-Free; Fig. S1a,c.3). Lacking dominant hubs, ER networks distribute dynamical variance evenly, preventing the "shadowing" effect of master regulators and allowing the algorithm to precisely pinpoint individual edges. Within this regime, structural recovery peaked at intermediate edge probabilities ($p \approx 0.075 - 0.10$,

AUPRC Ratio up to 33× random; Fig. S1b); extreme sparsity lacked sufficient causal coupling, while excessive density ($p \geq 0.30$) induced intractable topological confounding via indirect pathways.

However, as dimensionality scaled ($N \to 100$), topological inference on ER networks degraded significantly faster than on Scale-Free baselines (Fig. S1b,c). Because the average ER degree scales as $k \approx pN$, maintaining constant p in large dimensions yields vastly denser networks (e.g., $k \approx 30$ for $N = 100, p = 0.30$), collapsing the AUPRC Ratio to 1.43. Consequently, structural resolution in high dimensions strictly requires the absolute number of causal pathways per node to remain sparse ($p = 0.05$). Unlike purely random ER networks, which become exceedingly difficult to untangle from homogeneous background noise at scale, real-world Scale-Free topologies provide the necessary dynamical hubs that act as structural anchors, allowing Bayesian inference to robustly scale to macroscopic complex systems.

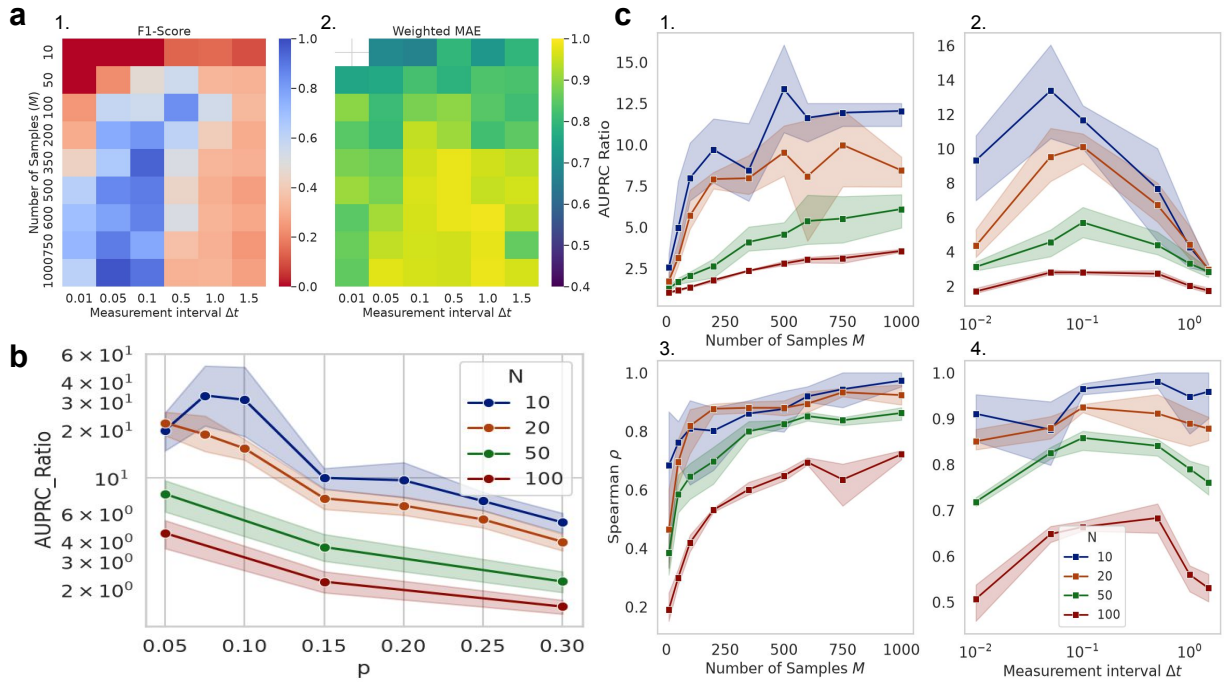


Figure S1: Grid search performance on Erdős-Rényi (ER) networks. a. Performance heatmaps as a function of the measurement interval Δt and the number of sample pairs M for an ER network of size $N = 10$ ($p = 0.15$) in the transient regime. **a.1** reports the F1-Score, showing topological recovery accuracy, while **a.2** reports the Weighted Mean Absolute Error (MAE), quantifying the reconstruction accuracy of the regulatory weights. **b.** Scalability with edge density: Area Under the Precision-Recall Curve Ratio (AUPRC Ratio) as a function of the edge probability p across network sizes $N \in \{10, 20, 50, 100\}$ at fixed sampling interval $\Delta t = 0.05$. For small networks ($N \leq 20$), intermediate densities ($p \approx 0.075$–0.10) maximize the performance relative to a random baseline, whereas for larger networks ($N = 50, 100$), performance decreases with p due to topological confounding from high average degrees ($k \approx pN$). **c.** Scaling of structural and parameter inference with sample size and temporal resolution for fixed edge probability $p = 0.15$ in the transient regime across network sizes $N \in \{10, 20, 50, 100\}$. **c.1** and **c.2** show the AUPRC Ratio as a function of sample size M (at fixed $\Delta t = 0.05$) and measurement interval Δt (at fixed $M = 500$), respectively. **c.3** and **c.4** show the Spearman rank correlation coefficient ρ of the inferred regulatory weights as a function of M (at fixed $\Delta t = 0.05$) and Δt (at fixed $M = 500$), respectively. Shaded bands represent standard deviations across independent realizations.